

# Restaurant Transaction Analysis and Expansion Recommendations for Foodie India

DOM207-DOM3007-DOM6401: MP1

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**Prepared for:** Foodie India Management

Data source: Foodie India proprietary restaurant transaction workbook

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# Section: Introduction

## Purpose

This project analyzes Foodie India's supplied restaurant transactions [1] to identify decision-relevant operating patterns and define reversible expansion pilots. It asks when higher-value dining occasions occur, how party size relates to per-head spend, what drives tip rate, and which operational tests are justified by the observed sample.

The Serve worksheet contains 365 records and seven supplied fields: Amount, Tip, Gender, Smoker, Day, Time, and Partysize. Findings are observational. The dataset contains no location, cost, margin, calendar date, capacity, footfall, or service-quality fields, so it cannot establish causation, profitability, site choice, or exposure-adjusted demand.

## Executive findings

- **The pooled day effect is confounded with service period.** Lunch represents 78.4% of valid Thursday records but only 16.0% of valid Sunday records. Day distributions differ within lunch ( $p=1.43e-05$ ) but not within dinner ( $p=0.314$ ).
- **Weekend lunch is the expansion opportunity to test.** Saturday and Sunday lunch has  $n=39$  and median bill 24.67, versus  $n=89$  and 16.27 for Thursday and Friday lunch ( $p=2.01e-06$ ).
- **Per-head spend declines as party size rises.** Spearman  $\rho=-0.197$  ( $p=2.24e-04$ ); median per-head spend is 8.22 for two-person parties and 6.43 for four-person parties.
- **Tip rate declines as bill amount rises.** After removing 3 probable tip-entry errors,  $\rho=-0.433$  ( $p=7.12e-18$ ). Tip rate does not differ by waiter gender, smoker status, service period, or day after false-discovery-rate control.

**Decision posture:** all recommendations are controlled pilots with stop rules. None is a permanent capacity, compensation, or site decision.

## Section: Data (pre)processing

### Cleaning and traceability

1. The script validates the seven required columns, copies the source data, and adds SourceRow. The workbook is never edited.
2. Obvious decimal-format defects in Amount and Tip are repaired; non-positive bills and missing or negative tips are excluded only from monetary analysis.
3. Unambiguous category variants are normalized. San and Sn map to Sunday; ambiguous S, SS, and SSS remain missing.
4. Integer party sizes 1-6 define per-head analysis. Per-head spend equals bill divided by party size.
5. Excel rows 303-310 are flagged as DuplicateBlock but retained. SourceRow 296 is separately flagged TipSuspect and retained. Tips exceeding three times a positive bill are flagged as TipEntryError.

### Data-quality audit

| Check                                                                            | Records | Treatment                                                                           |
|----------------------------------------------------------------------------------|---------|-------------------------------------------------------------------------------------|
| Source records                                                                   | 365     | Retained; source workbook unchanged                                                 |
| Exact duplicate-looking rows (7 exact; 8-row pasted block at Excel rows 303-310) | 7       | All retained; pasted copy flagged as DuplicateBlock                                 |
| Amount text formats repaired                                                     | 3       | Comma/internal hyphen converted to decimal                                          |
| Tip text formats repaired                                                        | 1       | Trailing backtick removed                                                           |
| Categorical values normalized                                                    | 65      | Only unambiguous variants mapped, including San/Sn to Sun                           |
| Ambiguous/missing day                                                            | 11      | Excluded only from day analysis                                                     |
| Ambiguous/missing time                                                           | 4       | Excluded only from time analysis                                                    |
| Ambiguous/missing smoker status                                                  | 9       | Excluded only from smoker analysis                                                  |
| Non-positive/unparseable amount                                                  | 1       | Excluded from monetary analysis                                                     |
| Missing/negative tip                                                             | 2       | Excluded from tip analysis                                                          |
| Analysis-ready amount/tip records                                                | 362     | Primary analytical base                                                             |
| Probable tip entry errors                                                        | 3       | Tip exceeds three times bill; excluded from tip-rate versus bill test               |
| Suspect original-block tip                                                       | 1       | SourceRow 296 has tip 13.55; retained in primary analysis and tested in sensitivity |
| Other flagged tip rates above 100%                                               | 4       | Retained; flagged but not asserted as errors                                        |
| Core party-size records (1-6)                                                    | 345     | Used for party-size and per-head analysis                                           |
| Pasted-block sensitivity sample                                                  | 354     | Drops SourceRow 303-310 only; primary conclusions re-estimated                      |

Counts are not additive because one row may meet more than one condition.

## Section: Data (pre)processing - anomaly review

### Pasted-block finding

Excel rows 295-302 reappear as rows 303-310. The copy is the cleaner version: original row 296 contains the suspect tip 13.55 (about 38% of a 35.60 bill), whereas copy row 304 contains 3.55. The seven exact duplicates comprise 5 exact rows inside the pasted block plus 2 unrelated exact pairs at Excel rows 198/199 and 254/316; 3 additional block rows differ slightly. Without a transaction ID, all records remain in the primary analysis.

Dropping the eight copied rows gives  $n=354$ . The fourth sensitivity also drops TipSuspect. Pooled day and correlation conclusions remain unchanged across all four scenarios.

| Scenario                           | n   | Pooled day p | Bill-tip rho | Bill-tip-rate rho |
|------------------------------------|-----|--------------|--------------|-------------------|
| Baseline                           | 362 | 0.003        | 0.639        | -0.422            |
| Drop DuplicateBlock                | 354 | 0.003        | 0.643        | -0.413            |
| Drop TipEntryError                 | 359 | 0.003        | 0.643        | -0.433            |
| Drop DuplicateBlock and TipSuspect | 353 | 0.003        | 0.640        | -0.423            |

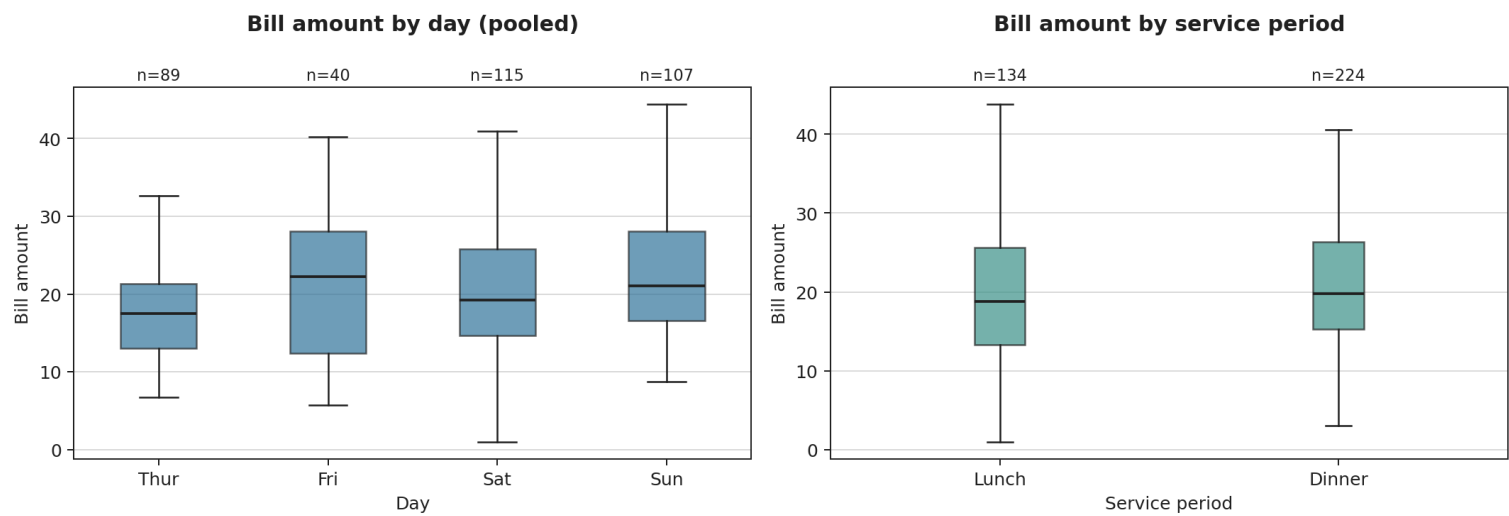
### Probable tip-entry errors

Three recorded tips exceed three times their positive bill and are probable decimal-loss errors. They remain in the audit file but are excluded from the primary bill-versus-tip-rate test and Figure 4.

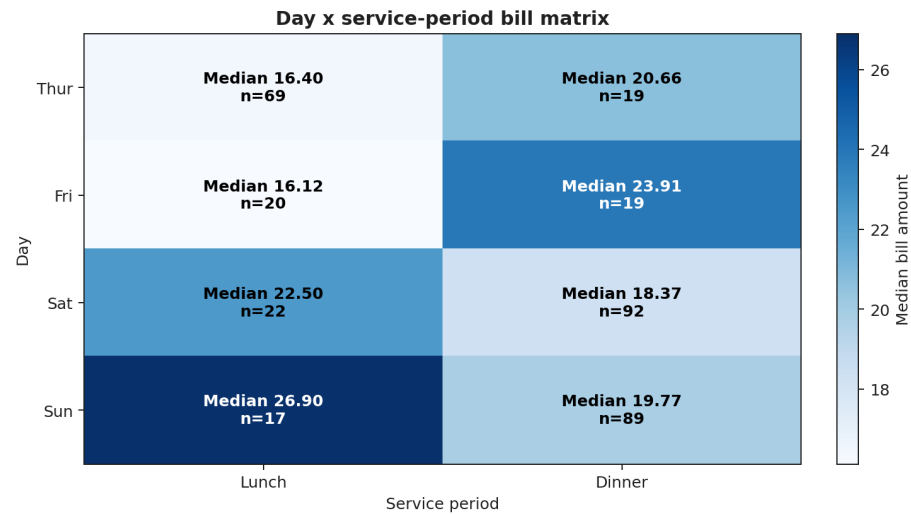
| Excel row | Bill  | Recorded tip | Recorded rate |
|-----------|-------|--------------|---------------|
| 91        | 21.16 | 300.00       | 1,417.8%      |
| 105       | 22.42 | 3487.00      | 15,553.1%     |
| 232       | 22.12 | 288.00       | 1,302.0%      |

Another 4 rates lie between 100% and 300%. They remain flagged but are not asserted to be errors. Restricting the association to rates at or below 100% gives  $n=355$ ,  $\rho=-0.429$ , and  $p=2.69e-17$ , so the negative relationship is robust.

# Section: Data visualization

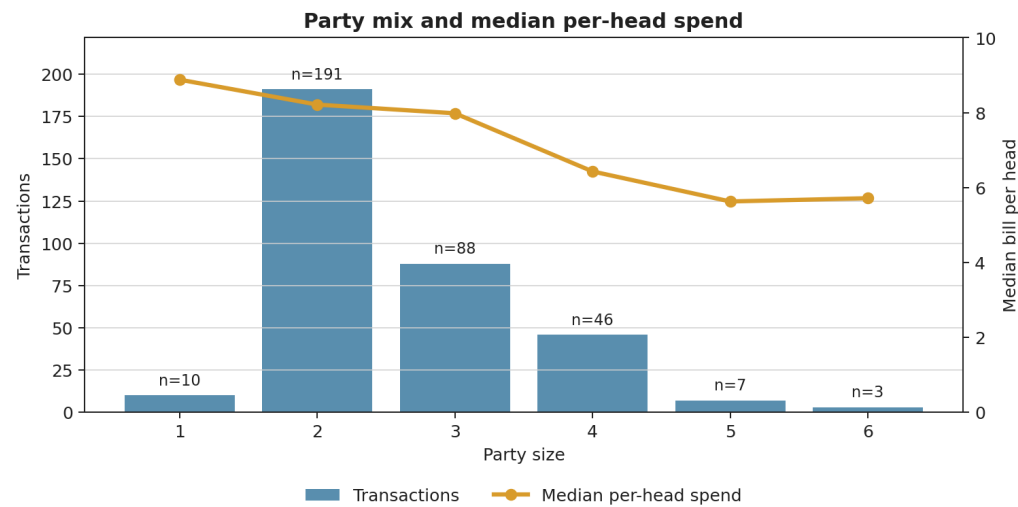


**Figure 1.** Boxplots compare pooled bill distributions by day and service period; n labels sit above the plotting areas. The pooled day panel uses n=351; the stratified day-by-service analysis uses n=347 because 4 valid-Day rows lack valid Time. For legibility, 14 day-panel and 11 service-panel plotting outliers are hidden under the 1.5 x IQR rule; all observations remain in tests.

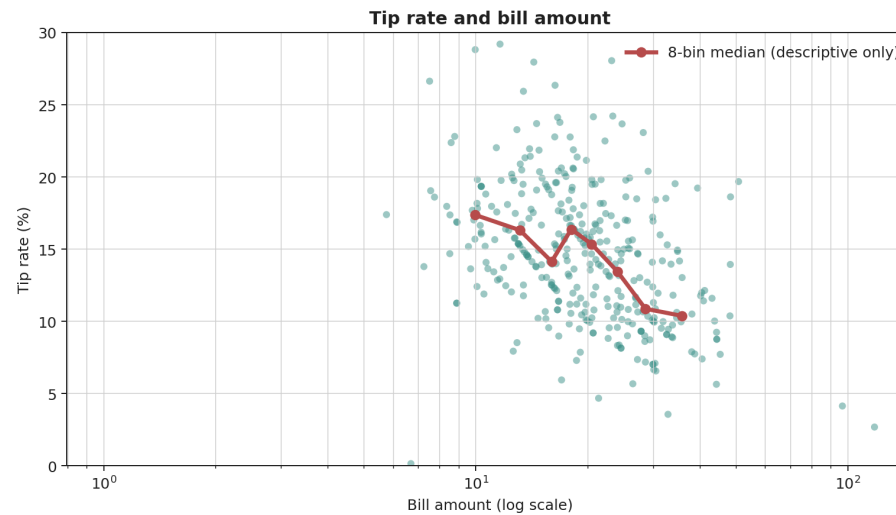


**Figure 2.** Heatmap of median bill and n for the n=347 records with valid Day and Time. Weekend lunch versus weekday lunch is the decision contrast. Lunch cells are small: Friday n=20, Saturday n=22, and Sunday n=17.

## Data visualization: decision measures



**Figure 3.** Bars show transaction n and the line shows median per-head spend for party sizes 1-6, with the per-head axis starting at zero. Party sizes 5 and 6 have only n=7 and n=3.



**Figure 4.** Scatterplot of tip rate versus bill on a logarithmic bill axis with an 8-bin median trend. The 3 probable entry errors are removed. The y-axis is capped at 30%; 7 valid points above the cap have rounded rates 33, 38, 65, 104, 112, 118, 166%. The trend is descriptive only.

## Section: Analysis

### Day findings must be interpreted within service period

The pooled day test and Figure 1 use every analysis-ready row with valid Day (n=351). The within-period tests and day-by-service table require valid Day and Time (n=347); the two bases differ because 4 valid-Day rows have missing or ambiguous Time.

The pooled day test is significant (Kruskal-Wallis  $H=13.841$ ,  $p=0.003$ , epsilon-squared=0.031). However, lunch exposure differs sharply by day:

| Day  | Valid n | Lunch n | Dinner n | Lunch share |
|------|---------|---------|----------|-------------|
| Thur | 88      | 69      | 19       | 78.4%       |
| Fri  | 39      | 20      | 19       | 51.3%       |
| Sat  | 114     | 22      | 92       | 19.3%       |
| Sun  | 106     | 17      | 89       | 16.0%       |

After stratification, lunch days differ ( $H=25.158$ ,  $p=1.43e-05$ , epsilon-squared=0.179), while dinner days do not ( $H=3.554$ ,  $p=0.314$ , epsilon-squared=0.003). Weekend lunch has median bill 24.67 (n=39) versus 16.27 (n=89) for weekday lunch; rank-biserial effect=0.529,  $p=2.01e-06$ . Thursday lunch and Friday lunch do not differ ( $p=0.709$ ). Thus the former pooled Thursday-Sunday interpretation is primarily a lunch/dinner composition effect.

For core-party lunch rows, weekend per-head spend has median 9.85 (n=36) versus 7.26 (n=83) on weekdays; C7 gives  $p=7.32e-05$  and rank-biserial effect=0.459. Mean party size is 2.69 versus 2.36, so the higher weekend per-head result is not explained by smaller parties.

### Methods and multiplicity

Rank-based procedures are used because monetary variables are skewed: Mann-Whitney U [2], Kruskal-Wallis [3], and Spearman rho [4]. Tests are two-sided. Benjamini-Hochberg adjustment [6] is applied separately to the eight primary and seven context tests; lunch post-hoc comparisons are adjusted separately. Effect sizes accompany p-values under ASA guidance [5], with the H-statistic epsilon-squared convention [9]. Calculations use SciPy [8].

| ID | Primary test                     | n   | Statistic  | Raw p    | BH p     | Effect | Decision         |
|----|----------------------------------|-----|------------|----------|----------|--------|------------------|
| H1 | Bill amount by day within lunch  | 128 | 25.158     | 1.43e-05 | 3.82e-05 | 0.179  | Reject H0        |
| H2 | Bill amount by day within dinner | 219 | 3.554      | 0.314    | 0.418    | 0.003  | Do not reject H0 |
| H3 | Weekend vs weekday lunch bill    | 128 | 2,654.000  | 2.01e-06 | 8.04e-06 | 0.529  | Reject H0        |
| H4 | Party size vs per-head spend     | 345 | -0.197     | 2.24e-04 | 4.48e-04 | -0.197 | Reject H0        |
| H5 | Bill amount vs tip rate          | 359 | -0.433     | 7.12e-18 | 5.70e-17 | -0.433 | Reject H0        |
| H6 | Tip rate by waiter gender        | 362 | 16,263.000 | 0.802    | 0.802    | 0.015  | Do not reject H0 |
| H7 | Tip rate by smoker status        | 353 | 16,249.500 | 0.190    | 0.304    | 0.082  | Do not reject H0 |
| H8 | Tip rate by service period       | 358 | 14,434.000 | 0.545    | 0.623    | -0.038 | Do not reject H0 |

# Analysis: commercial interpretation

## Per-head spend

| Party size | n   | Median bill | Median per head |
|------------|-----|-------------|-----------------|
| 1          | 10  | 8.88        | 8.88            |
| 2          | 191 | 16.43       | 8.22            |
| 3          | 88  | 23.95       | 7.98            |
| 4          | 46  | 25.73       | 6.43            |
| 5          | 7   | 28.15       | 5.63            |
| 6          | 3   | 34.30       | 5.72            |

Per-head spend declines with party size ( $\rho=-0.197$ ,  $p=2.24e-04$ ). Two-person median is 8.22, compared with 6.43 for four-person parties; the exploratory two-versus-four comparison gives  $p=4.85e-04$  and  $\text{effect}=0.332$ . The positive party-size-versus-total-bill correlation is context only and does not show that larger groups are more productive per seat.

## Tip rate

| Bill quartile | n  | Minimum bill | Maximum bill | Median tip rate |
|---------------|----|--------------|--------------|-----------------|
| Q1 (lowest)   | 90 | 1.01         | 14.31        | 16.9%           |
| Q2            | 90 | 14.48        | 19.10        | 15.8%           |
| Q3            | 89 | 19.12        | 25.71        | 14.3%           |
| Q4 (highest)  | 90 | 25.89        | 117.81       | 10.6%           |

Median tip rate falls from 16.9% in the lowest bill quartile to 10.6% in the highest. The primary association is  $\rho=-0.433$ ,  $p=7.12e-18$ . Tip rate does not differ by waiter gender ( $p=0.802$ ), smoker status ( $p=0.190$ ), service period ( $p=0.545$ ), or day ( $p=0.331$ ).

The decline is partly a **flat-dollar** tip effect: median tip is 2.87, and 144 of 359 tips (40.1%) are multiples of 0.50. Tip rate is negatively associated with per-head spend ( $\rho=-0.455$ ,  $p=7.61e-19$ ) but not clearly with party size ( $\rho=-0.095$ ,  $p=0.080$ ). Therefore the bill-rate pattern does not show that larger spenders tip less generously. Bill and tip amount  $\rho=0.639$  remains mechanical context.

## Limitations

- Sampling frame, collection dates, transaction independence, and current representativeness are unknown.
- Lunch cells are small: Friday  $n=20$ , Saturday  $n=22$ , Sunday  $n=17$ .
- Party size may include children or shared dishes; per-head spend is an approximation of per-diner spend.
- No cost, margin, discount, labour, capacity, location, or footfall fields exist.
- No transaction ID exists; the pasted block cannot be conclusively classified.
- All results are observational and do not establish causation.

**Interpretation:** service-period composition prevents a standalone Sunday or Thursday action. Pilots must compare like-for-like shifts.



## Section: Discussions and Recommendations

### R1. Weekend lunch service pilot

**Justification:** Saturday and Sunday lunch has median bill 24.67 (n=39). Core-party weekend lunch also has higher median per-head spend (9.85, n=36) than weekday lunch (7.26, n=83), with  $p=7.32e-05$ ; mean party size is larger, not smaller (2.69 versus 2.36).

**Action:** test an extended or added weekend lunch window on matched dates with comparable untreated shifts.

**Success metric:** incremental covers, labour hours, gross profit per available seat-hour, median bill, and complaints versus control.

**Stop rule:** stop if gross profit per available seat-hour does not improve after added labour, or complaints materially worsen.

### R2. Thursday lunch value-per-head pilot

**Justification:** Thursday lunch has the largest lunch cell (n=69) and lowest observed lunch per-head median (6.85, n=63 valid core-party rows).

**Action:** test one transparent add-on or meal-upgrade prompt on matched Thursday lunch shifts; do not scale on volume alone.

**Success metric:** per-head spend, attachment rate, gross profit per seat-hour, and complaints.

**Stop rule:** stop if per-head spend and gross profit per seat-hour do not improve together or complaints increase.

### R3. Small-party and flexible-table pilot

**Justification:** median per-head spend is 8.22 for two-person and 6.43 for four-person parties; the overall association is negative. Large-party bundles are not supported.

**Action:** favour two-top tables that pair flexibly. Treat any three-to-four-person add-on as a measured experiment.

**Success metric:** per-head spend, seat utilization, turn time, and gross profit per seat-hour.

**Stop rule:** revert if utilization or gross profit per seat-hour falls without a guest-service benefit.

# Recommendations: controls and boundaries

## R4. Tip-policy control

**Justification:** tip rate declines as bills rise; no adjusted difference is established by waiter gender, smoker status, service period, or day. Hospitality evidence warns that raw tip size is a weak service-quality measure [7].

**Action:** do not rank, schedule, or pay staff using raw tips. Evaluate any service coaching with guest feedback and rate-based measures.

**Success metric:** guest ratings, complaints, tip-rate distribution, and staff dispersion after controlling for bill mix.

**Stop rule:** withdraw incentives that create demographic segmentation, unstable rankings, or no guest-outcome improvement.

## R5. Data-capture prerequisite

**Justification:** the workbook cannot distinguish sites, dates, repeat records, costs, margins, capacity, or exposure.

**Action:** add timestamp, transaction ID, location, menu/category, discount, direct-cost, labour-hour, table-capacity, and available-seat-hour fields.

**Success metric:** complete identifiers and exposure/cost fields for every pilot transaction, with reconciliation checks.

**Stop rule:** do not approve permanent expansion while transaction identity, cost, and capacity exposure remain unavailable.

## Actions not supported by the data

- No city, market, or site recommendation.
- No permanent capacity addition based on transaction counts.
- No four-person bundle as a default proposition.
- No staffing or compensation decision based on raw tips, waiter gender, or smoker status.
- No causal claim about spend, tips, satisfaction, or retention.
- No profit or return-on-investment forecast.

## Section: References

- [1] Foodie India. (n.d.). *Restaurant.xlsx* [Proprietary transaction dataset] and *DOM207-DOM3007-DOM6401: MP1* [Project specification]. Supplied course files; accessed 21 September 2026.
- [2] Mann, H. B., and Whitney, D. R. (1947). "On a Test of Whether One of Two Random Variables Is Stochastically Larger than the Other." *Annals of Mathematical Statistics*, 18(1), 50-60. <https://doi.org/10.1214/aoms/1177730491>.
- [3] Kruskal, W. H., and Wallis, W. A. (1952). "Use of Ranks in One-Criterion Variance Analysis." *Journal of the American Statistical Association*, 47(260), 583-621. <https://doi.org/10.1080/01621459.1952.10483441>.
- [4] Spearman, C. (1904). "The Proof and Measurement of Association between Two Things." *The American Journal of Psychology*, 15(1), 72-101. <https://doi.org/10.2307/1412159>.
- [5] Wasserstein, R. L., and Lazar, N. A. (2016). "The ASA's Statement on p-Values: Context, Process, and Purpose." *The American Statistician*, 70(2), 129-133. <https://doi.org/10.1080/00031305.2016.1154108>.
- [6] Benjamini, Y., and Hochberg, Y. (1995). "Controlling the False Discovery Rate: A Practical and Powerful Approach to Multiple Testing." *Journal of the Royal Statistical Society: Series B*, 57(1), 289-300. <https://doi.org/10.1111/j.2517-6161.1995.tb02031.x>.
- [7] Lynn, M., and McCall, M. (2000). "Gratitude and Gratitude: A Meta-Analysis of Research on the Service-Tipping Relationship." *Journal of Socio-Economics*, 29(2), 203-214. [https://doi.org/10.1016/S1053-5357\(00\)00062-7](https://doi.org/10.1016/S1053-5357(00)00062-7).
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- [9] Kassambara, A. (2023). *rstatix: Kruskal-Wallis Effect Size*. R package documentation. Formula  $\eta^2[H] = (H - k + 1)/(n - k)$ . [https://search.r-project.org/CRAN/refmans/rstatix/html/kruskal\\_effsize.html](https://search.r-project.org/CRAN/refmans/rstatix/html/kruskal_effsize.html); accessed 21 September 2026.

## Reproducibility statement

Run `python analysis.py` to regenerate cleaned data, audit tables, tests, sensitivities, and figures. Then run `python verify.py` to check source-row preservation, flags, test families, figures, report content, and landscape layout.